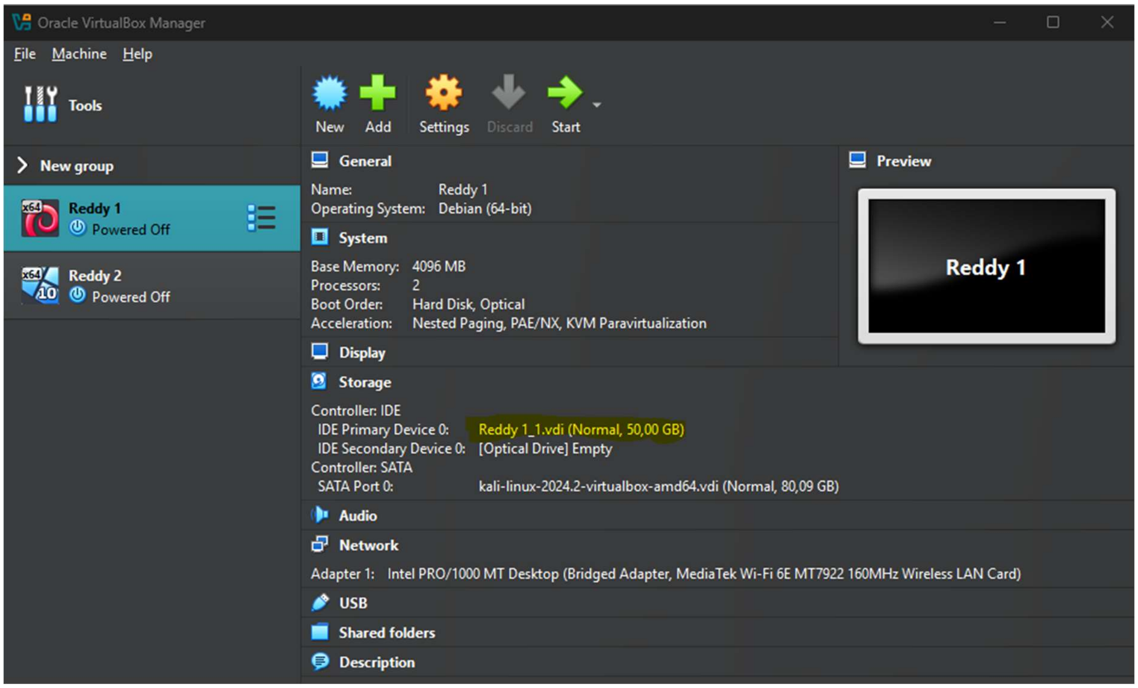
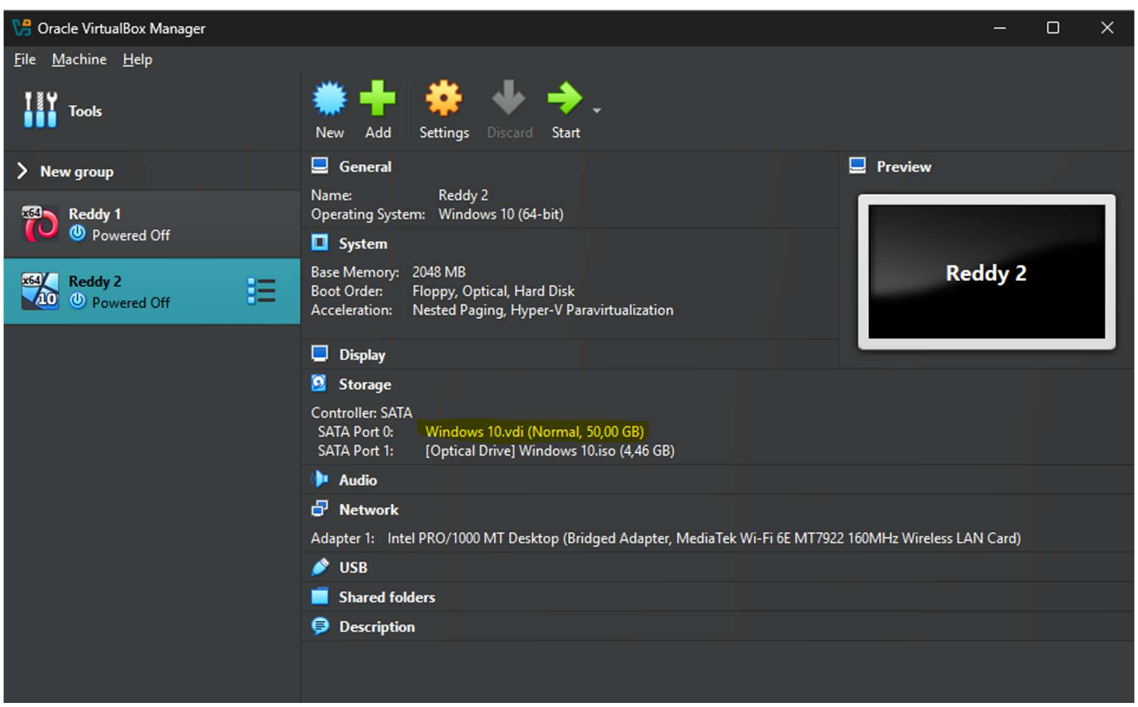
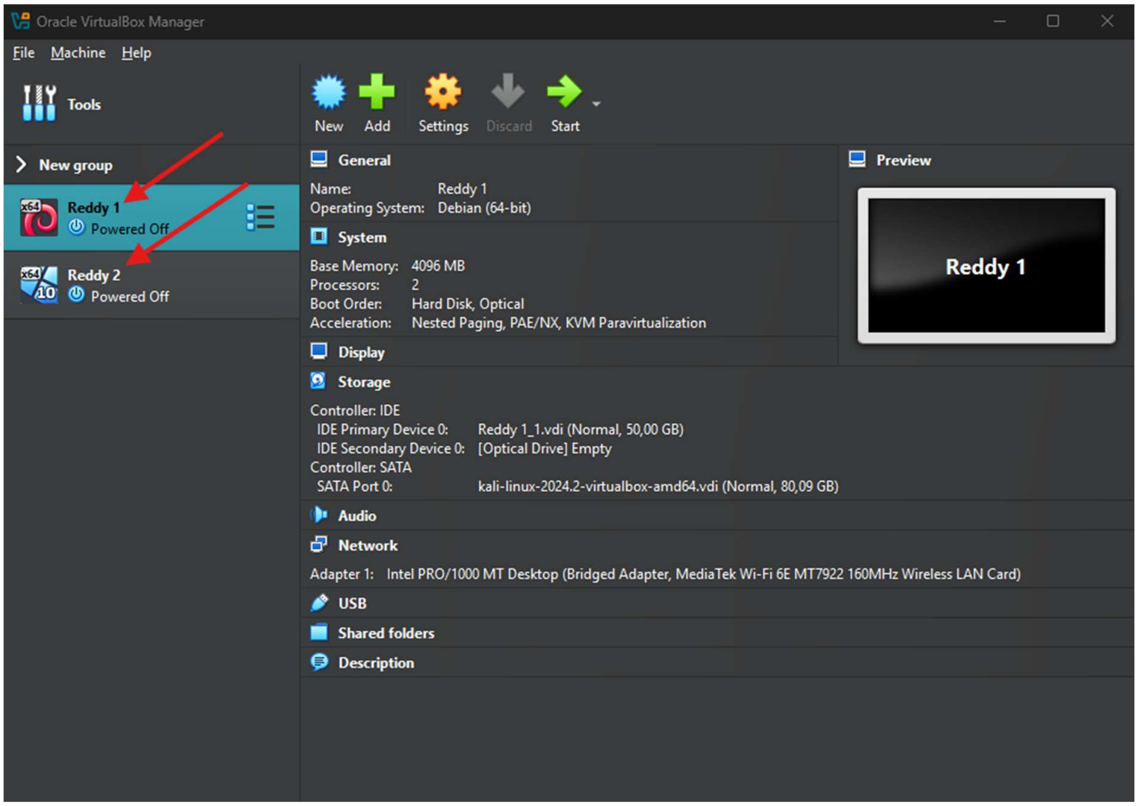
1. Create two virtual machines using Oracle VM with at least the following specifics: RAM: Use the RAM size of your choice based on the amount of RAM you have on your laptop/desktop, 50GB HDD.

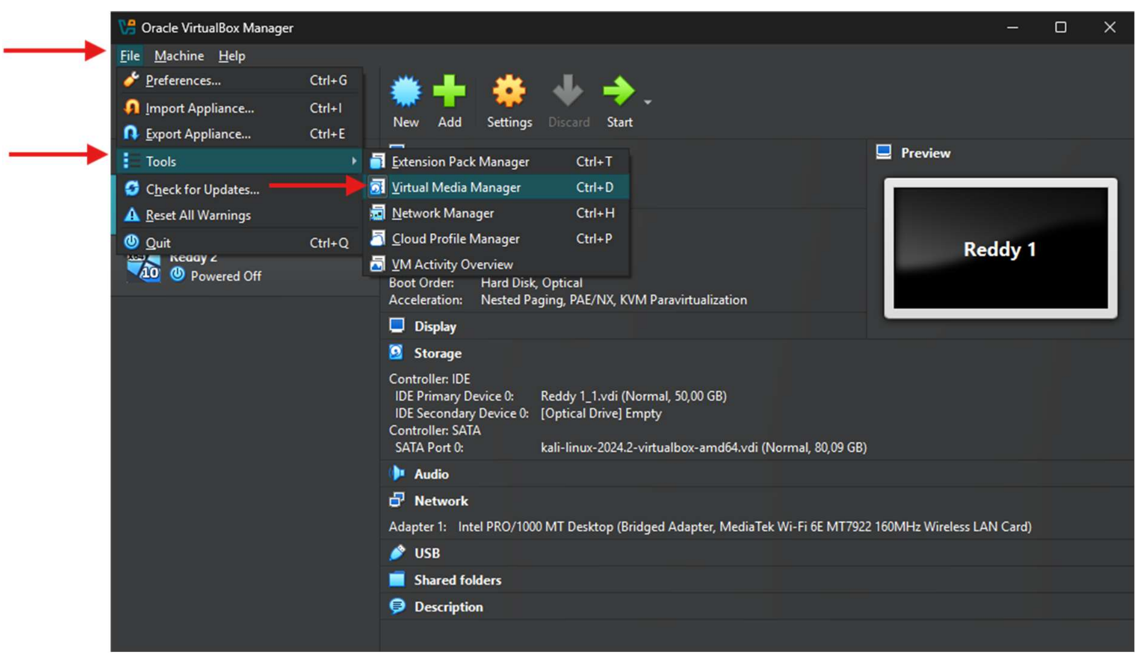




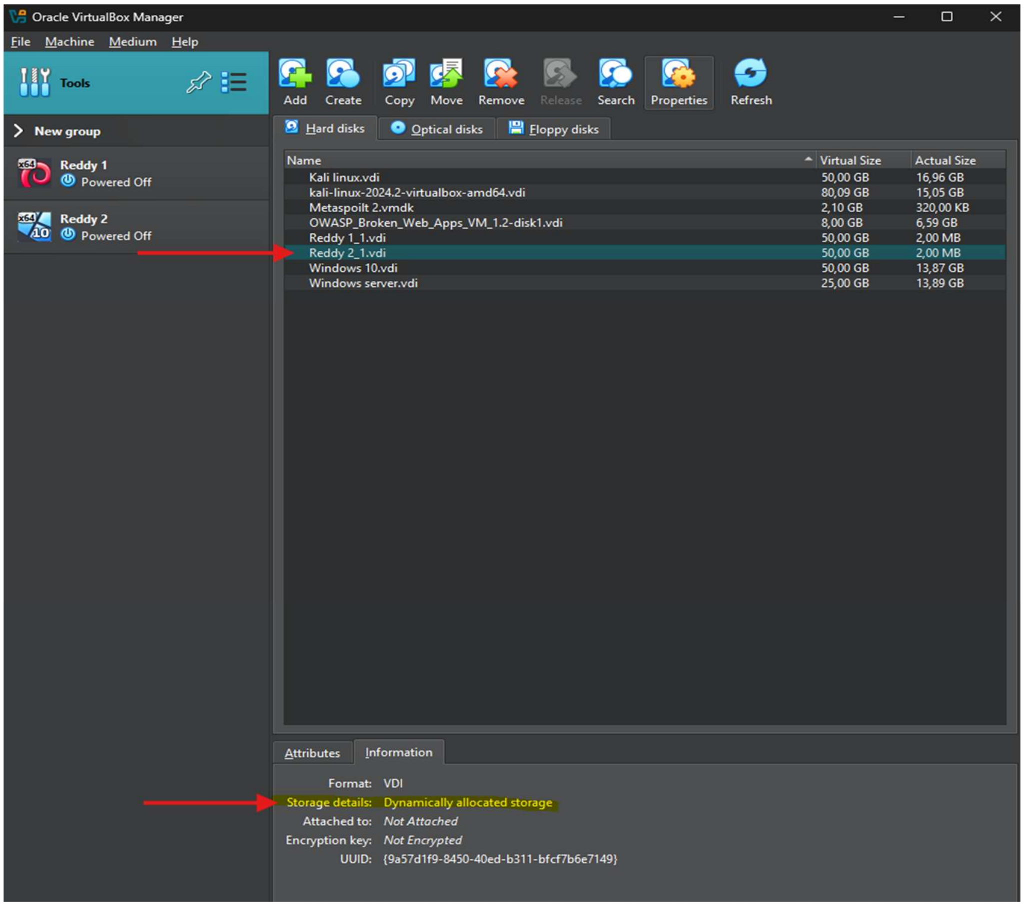
As naming convention, please use “**your surname**” instead of “VM”

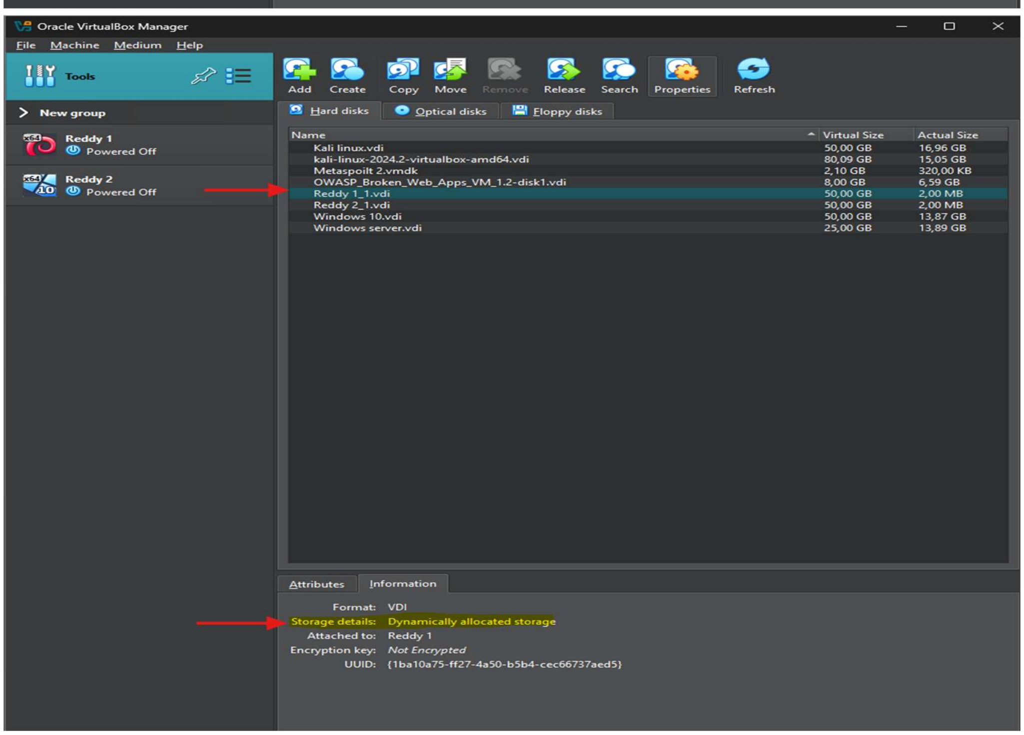


2. Using screenshots, show how you will configure the storage size of your VMs to grow as you add up more files on the hard drives.

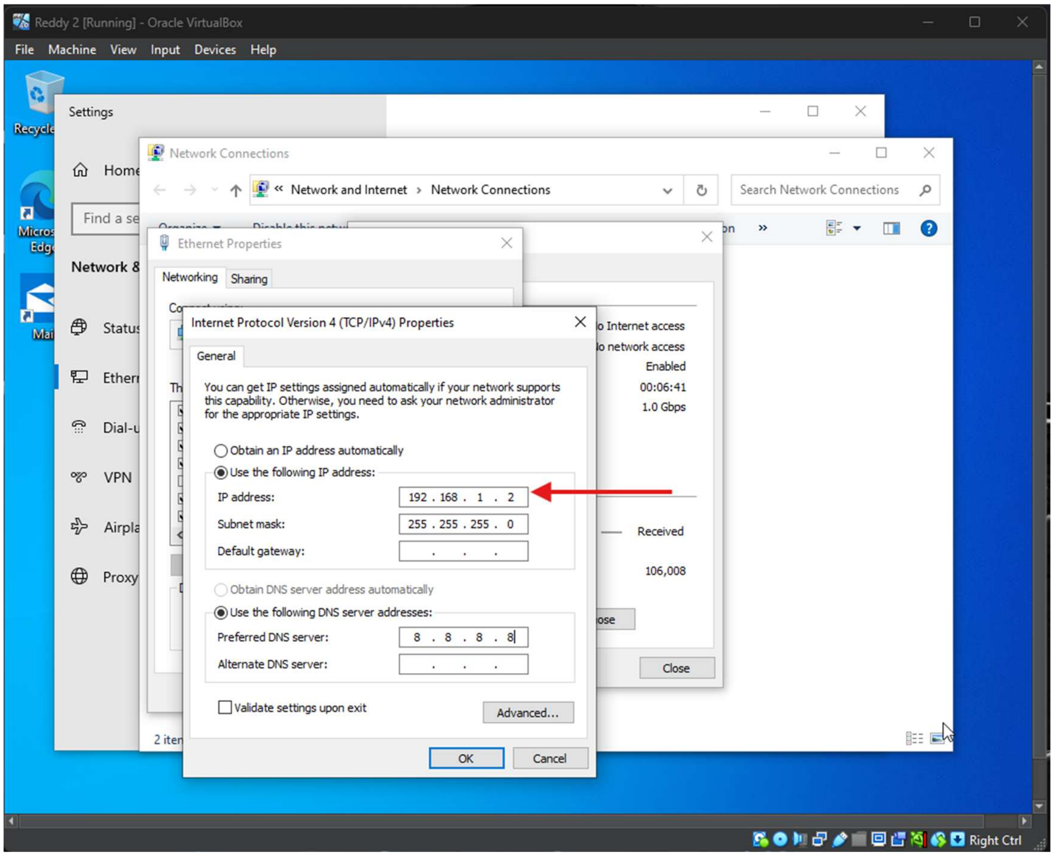


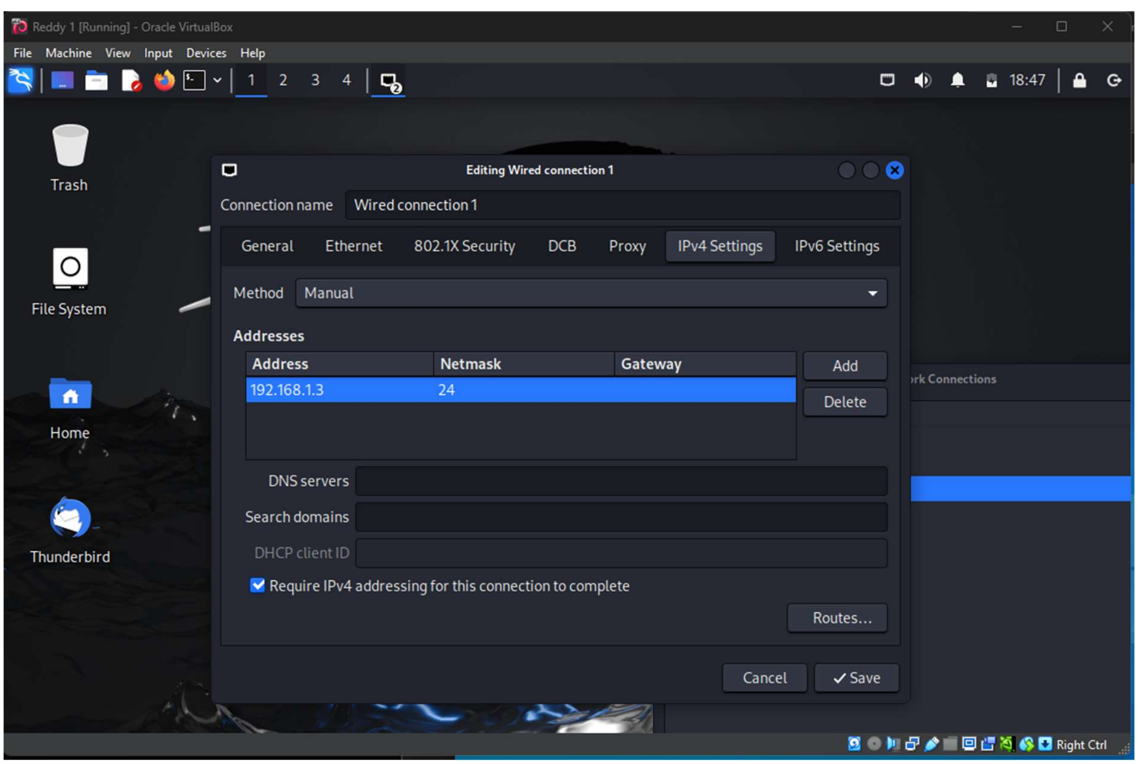
Set storge to Dynamic so space used on physical disk and can grow as needed.

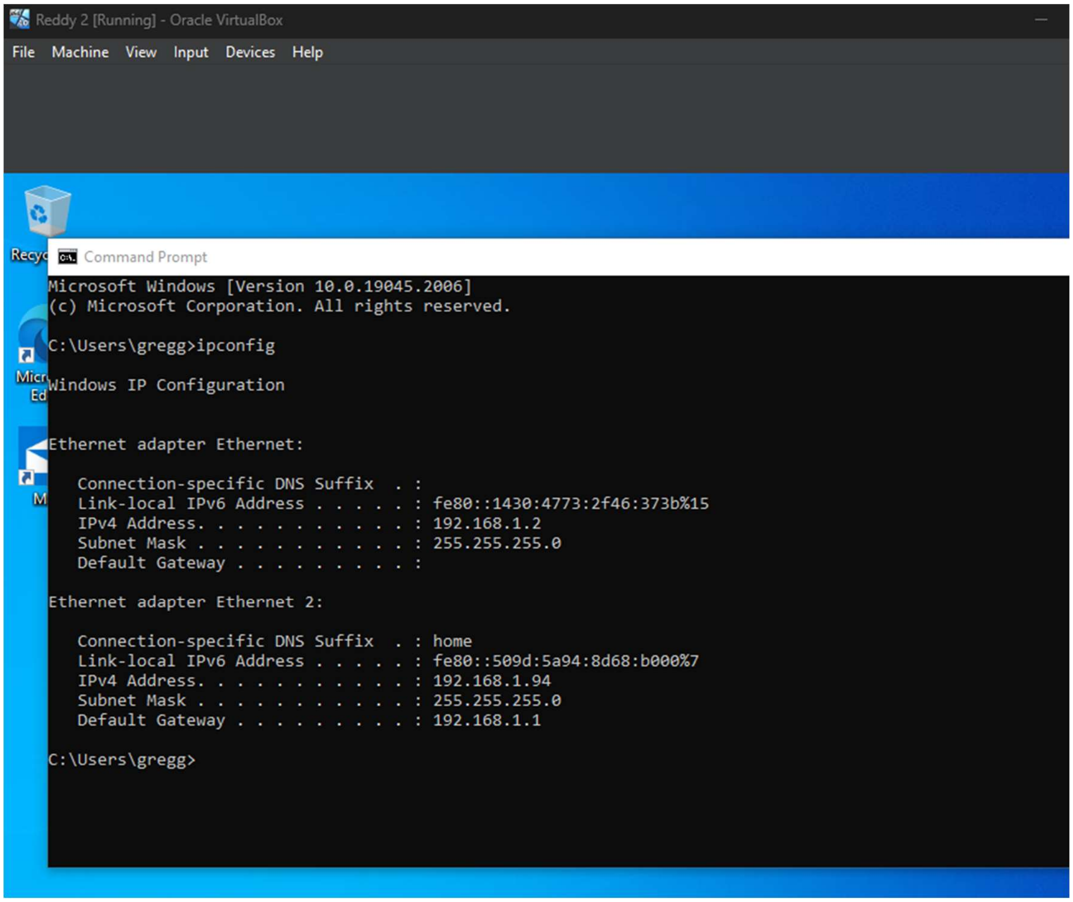


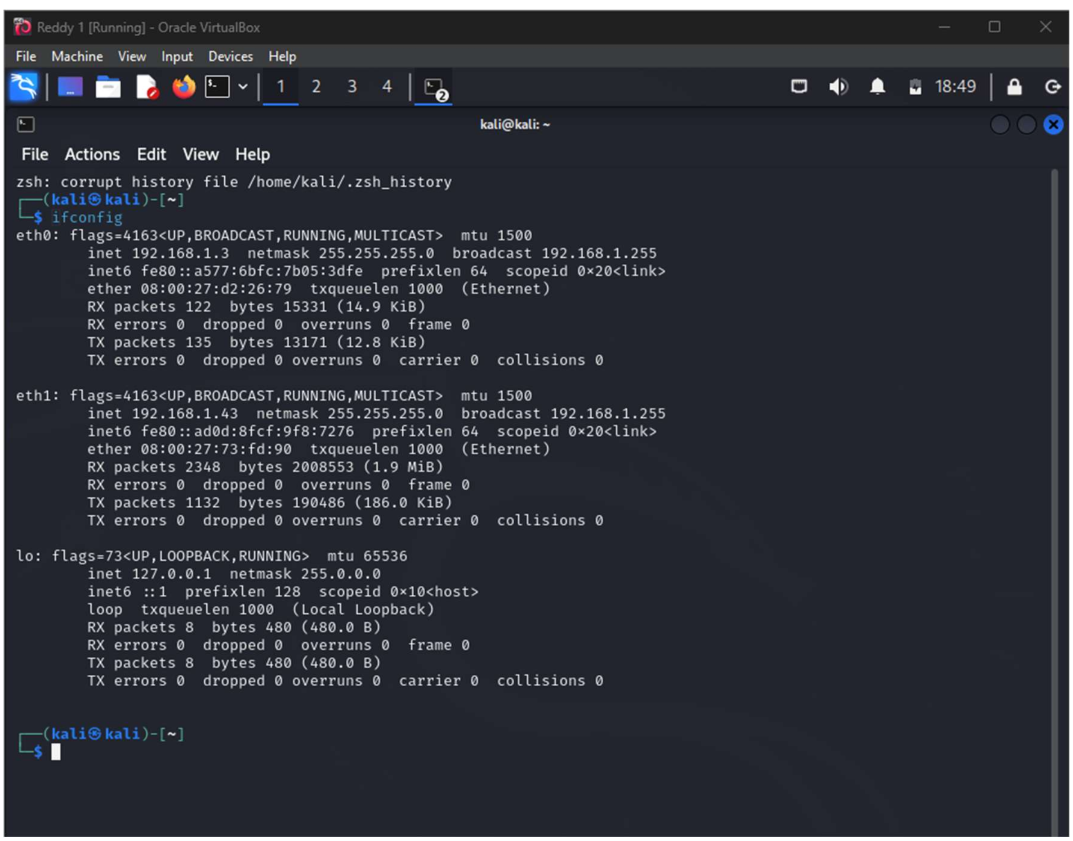


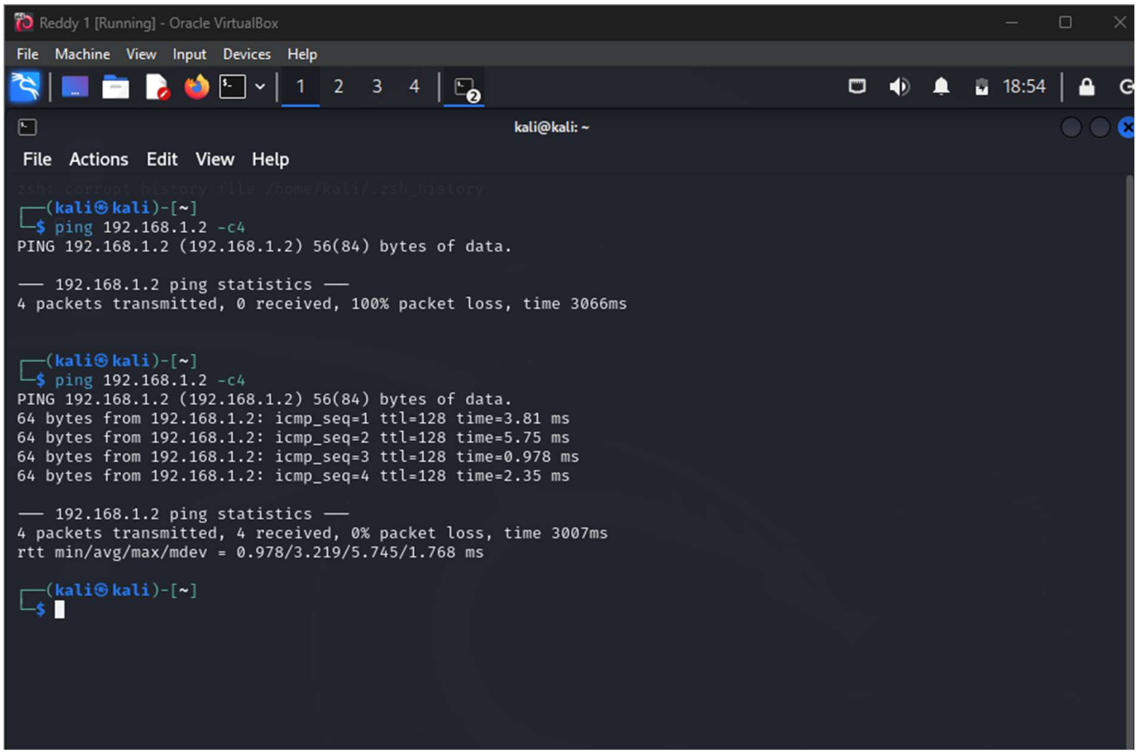
3. Install and configure respectively Windows and Kali-Linux on these two VMs.



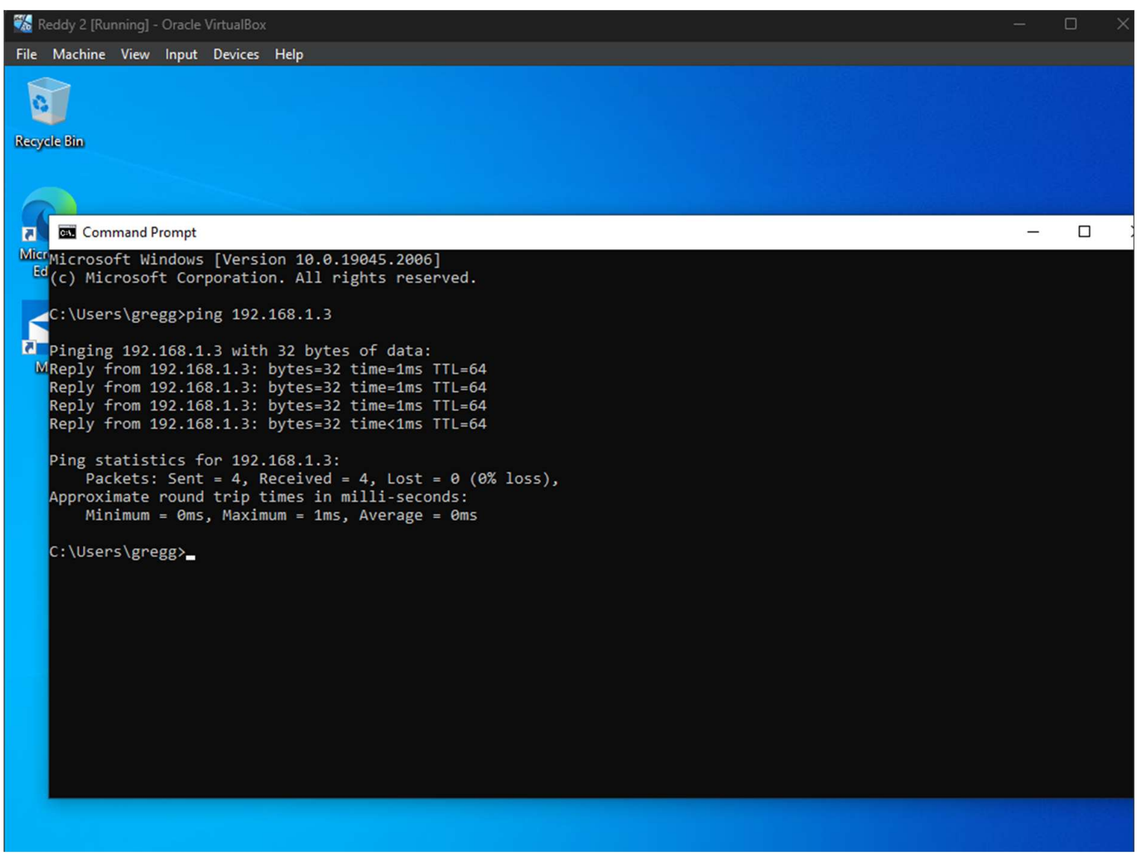



Reddy 2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Command Prompt
Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.
C:\Users\gregg>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::1430:4773:2f46:373b%15
IPv4 Address. . . . . : 192.168.1.2
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . :
Ethernet adapter Ethernet 2:
Connection-specific DNS Suffix . : home
Link-local IPv6 Address . . . . . : fe80::509d:5a94:8d68:b000%7
IPv4 Address. . . . . : 192.168.1.94
Subnet Mask . . . . : 255.255.255.0
Default Gateway . . . . : 192.168.1.1
C:\Users\gregg>


Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
18:49
kali@kali: ~
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.1.3 netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::a577:6bfc:7b05:3dfe prefixlen 64 scopeid 0x20<link>
ether 08:00:27:d2:26:79 txqueuelen 1000 (Ethernet)
RX packets 122 bytes 15331 (14.9 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 135 bytes 13171 (12.8 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.1.43 netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::ad0d:8fcf:9f8:7276 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:73:fd:90 txqueuelen 1000 (Ethernet)
RX packets 2348 bytes 2008553 (1.9 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1132 bytes 190486 (186.0 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 8 bytes 480 (480.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 8 bytes 480 (480.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$

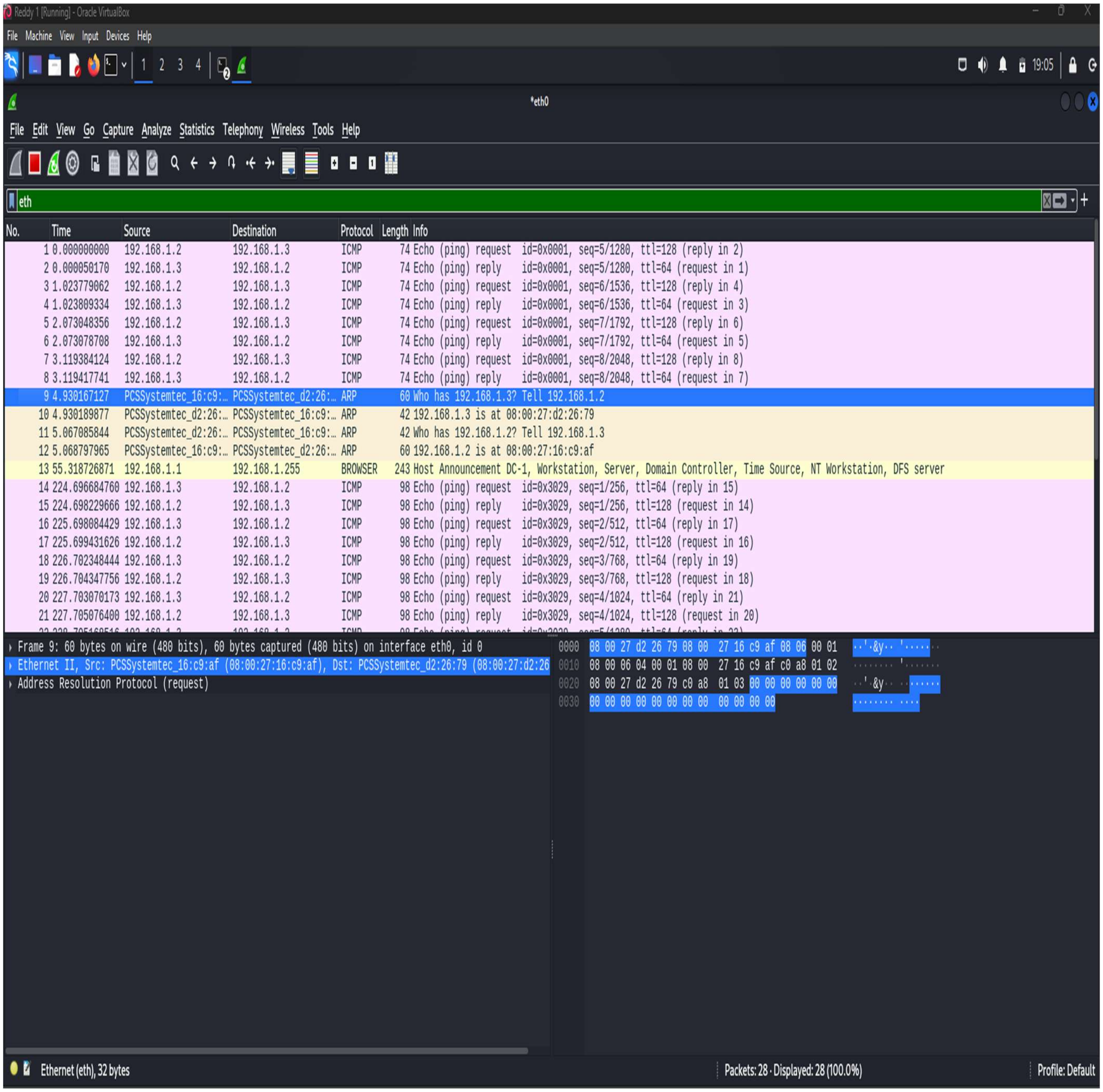


Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
18:54
kali@kali: ~
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ ping 192.168.1.2 -c4
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
--- 192.168.1.2 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3066ms
┌──(kali㉿kali)-[~]
└─$ ping 192.168.1.2 -c4
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=128 time=3.81 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=128 time=5.75 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=128 time=0.978 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=128 time=2.35 ms
--- 192.168.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 0.978/3.219/5.745/1.768 ms
┌──(kali㉿kali)-[~]
└─$

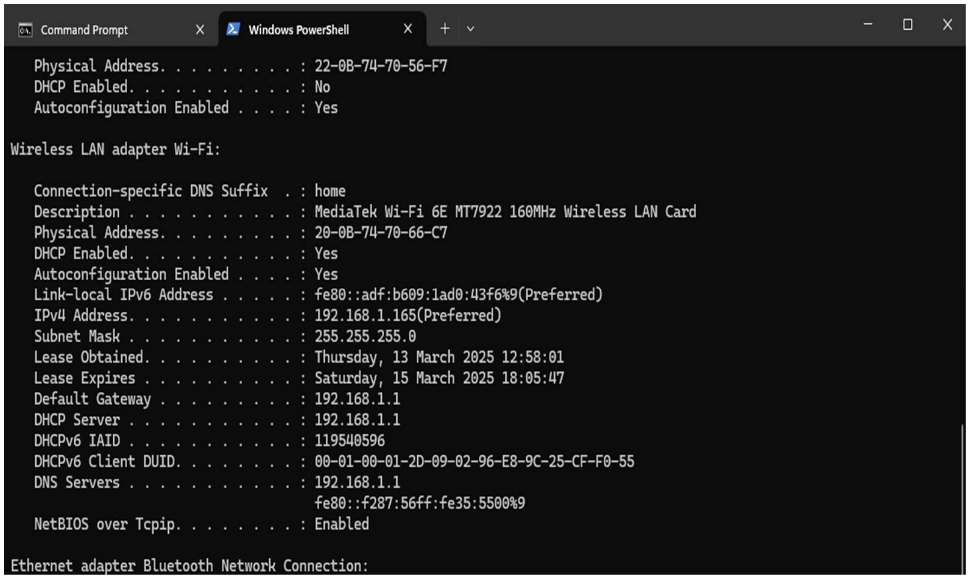


Reddy 2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Recycle Bin
Command Prompt
Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.
C:\Users\gregg>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=1ms TTL=64
Reply from 192.168.1.3: bytes=32 time=1ms TTL=64
Reply from 192.168.1.3: bytes=32 time=1ms TTL=64
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\gregg>

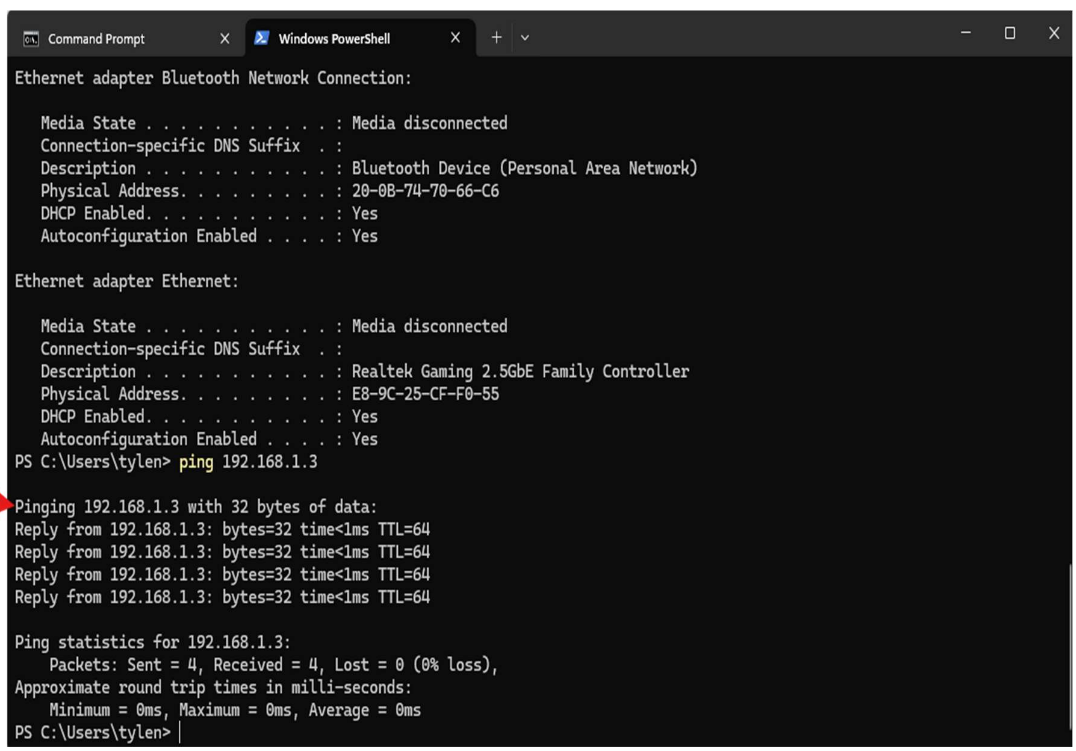
Set up a Windows virtual machine (VM) and a Kali Linux VM on the same network, simulate network traffic, and analyse it using Wireshark.



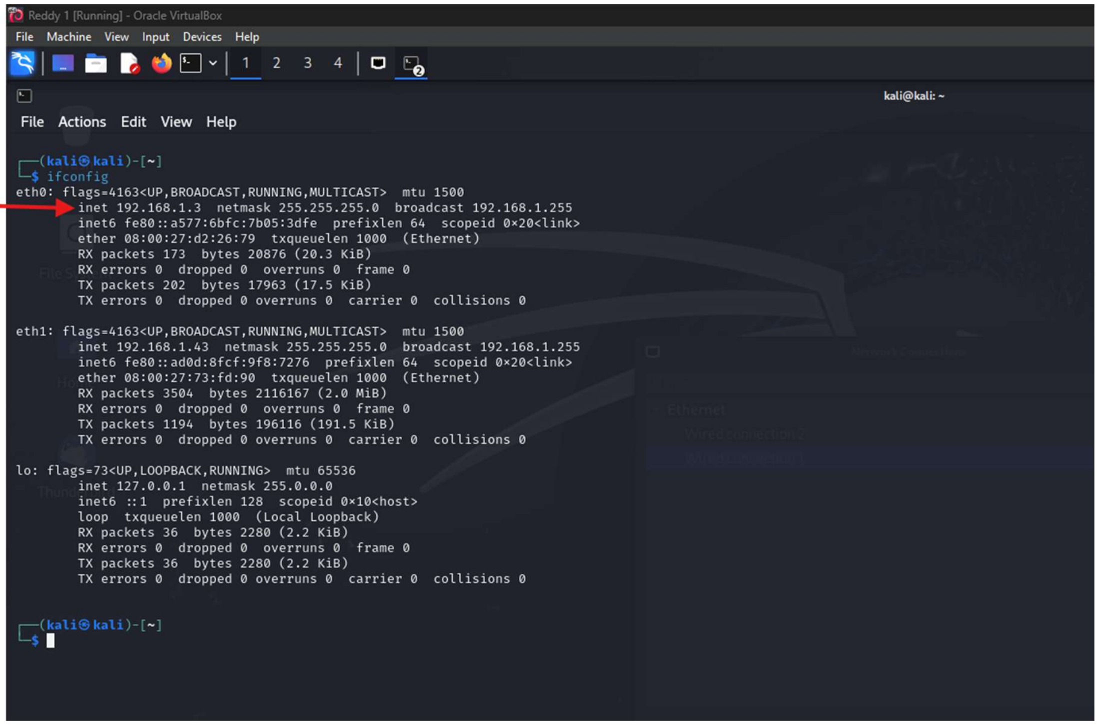
1. Make sure the host machine, and the VM_1 machine both are in the same network and can connect to the internet. As proof, provide network interface details by running appropriate commands from the two machines respectively.

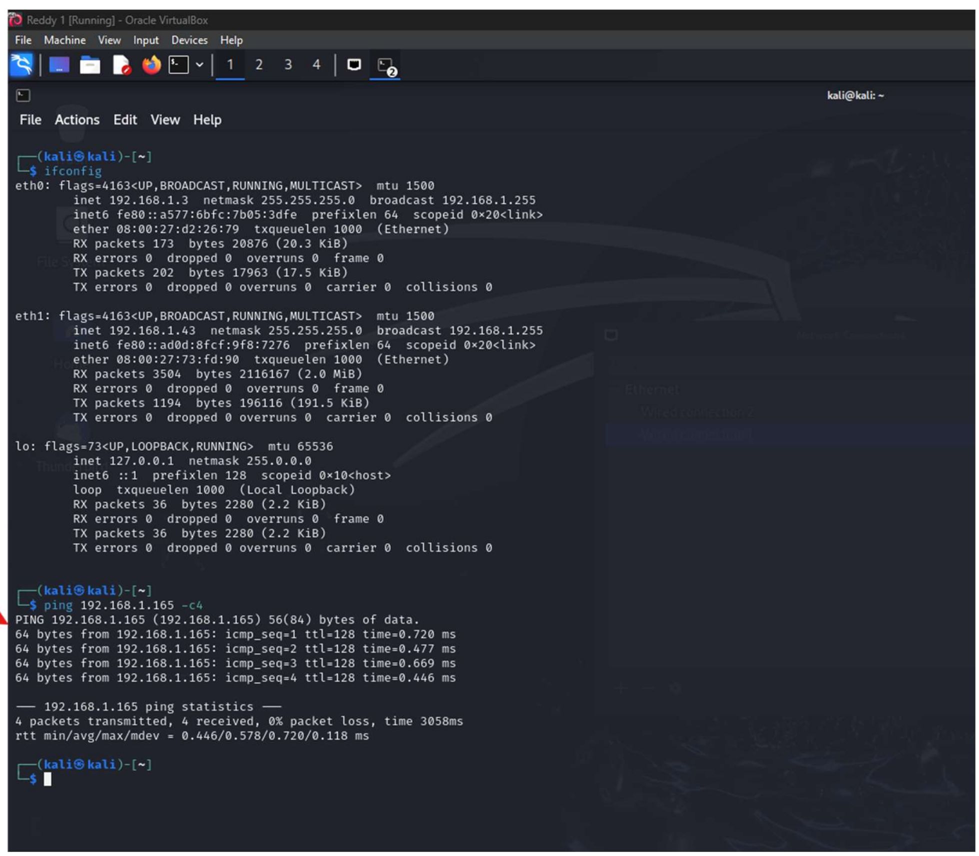


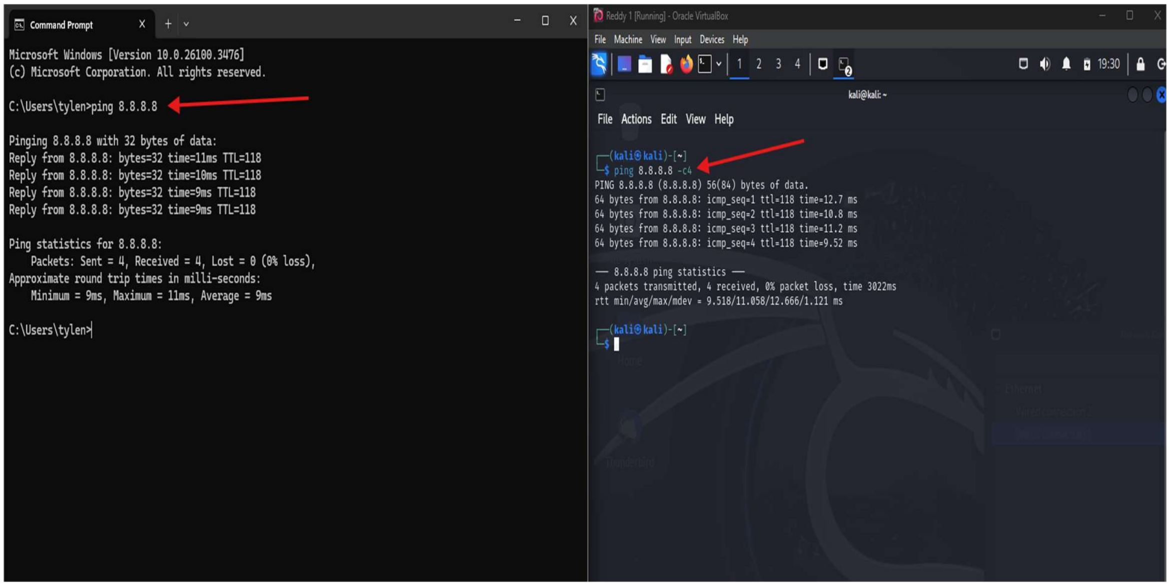
```
Command Prompt    Windows PowerShell
   Physical Address. . . . . . . . . : 22-0B-74-70-56-F7
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : home
   Description . . . . . . . . . . . : MediaTek Wi-Fi 6E MT7922 160MHz Wireless LAN Card
   Physical Address. . . . . . . . . : 20-0B-74-70-66-C7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::adf:b609:1ad0:43f6%9(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.165(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, 13 March 2025 12:58:01
   Lease Expires . . . . . . . . . . : Saturday, 15 March 2025 18:05:47
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 119540596
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-09-02-96-E8-9C-25-CF-F0-55
   DNS Servers . . . . . . . . . . . : 192.168.1.1
                                       fe80::f287:56ff:fe35:5500%9
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:
```



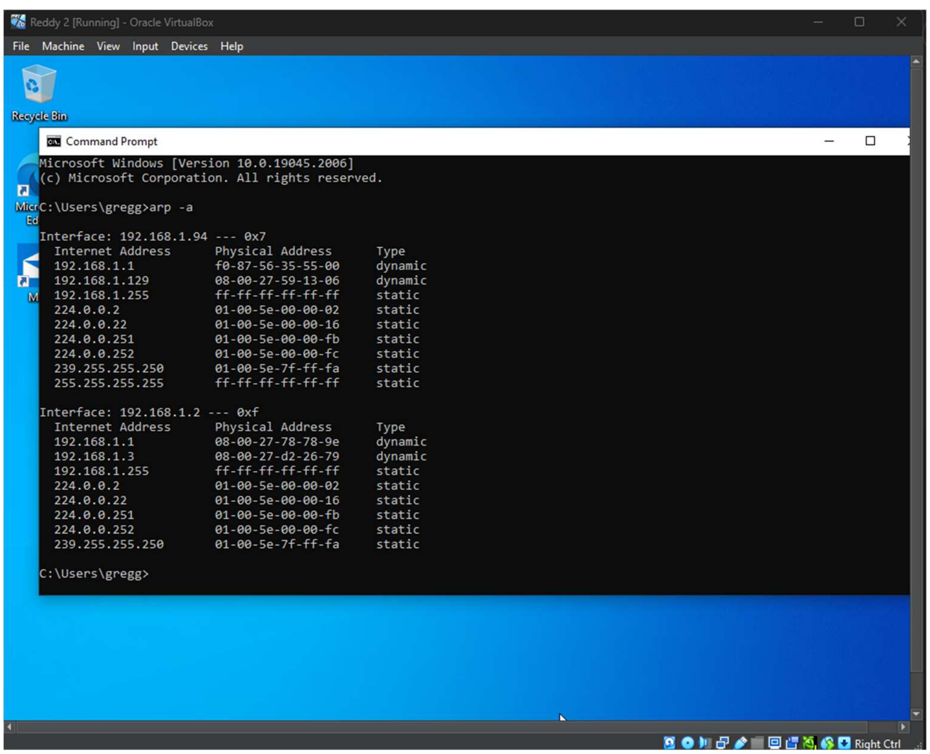
```
Command Prompt    Windows PowerShell
Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 20-0B-74-70-66-C6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming 2.5GbE Family Controller
   Physical Address. . . . . . . . . : E8-9C-25-CF-F0-55
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
PS C:\Users\tylen> ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\tylen>
```

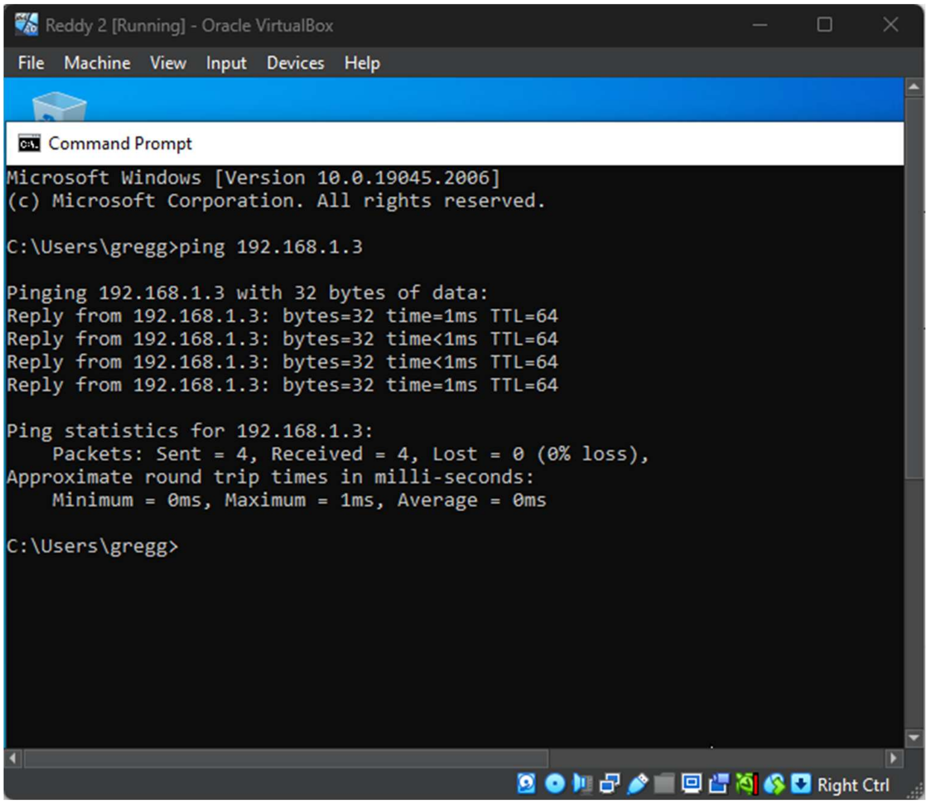

Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
File Actions Edit View Help
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.1.3 netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::a577:6bfc:7b05:3dfe prefixlen 64 scopeid 0×20<link>
ether 08:00:27:d2:26:79 txqueuelen 1000 (Ethernet)
RX packets 173 bytes 20876 (20.3 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 202 bytes 17963 (17.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.1.43 netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::ad0d:8fcf:9f8:7276 prefixlen 64 scopeid 0×20<link>
ether 08:00:27:73:fd:90 txqueuelen 1000 (Ethernet)
RX packets 3504 bytes 2116167 (2.0 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1194 bytes 196116 (191.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0×10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 36 bytes 2280 (2.2 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 36 bytes 2280 (2.2 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$

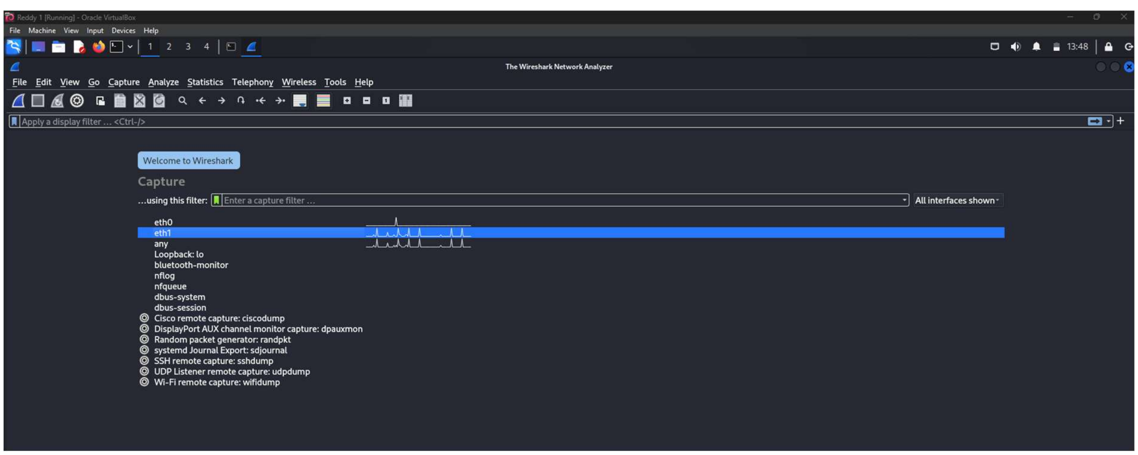

Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: ~
File Actions Edit View Help
(kali㉿kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.1.3 netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::a577:6bfc:7b05:3dfe prefixlen 64 scopeid 0×20<link>
ether 08:00:27:d2:26:79 txqueuelen 1000 (Ethernet)
RX packets 173 bytes 20876 (20.3 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 202 bytes 17963 (17.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.1.43 netmask 255.255.255.0 broadcast 192.168.1.255
inet6 fe80::ad0d:8fcf:9f8:7276 prefixlen 64 scopeid 0×20<link>
ether 08:00:27:73:fd:90 txqueuelen 1000 (Ethernet)
RX packets 3504 bytes 2116167 (2.0 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 1194 bytes 196116 (191.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0×10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 36 bytes 2280 (2.2 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 36 bytes 2280 (2.2 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
(kali㉿kali)-[~]
$ ping 192.168.1.165 -c4
PING 192.168.1.165 (192.168.1.165) 56(84) bytes of data.
64 bytes from 192.168.1.165: icmp_seq=1 ttl=128 time=0.720 ms
64 bytes from 192.168.1.165: icmp_seq=2 ttl=128 time=0.477 ms
64 bytes from 192.168.1.165: icmp_seq=3 ttl=128 time=0.669 ms
64 bytes from 192.168.1.165: icmp_seq=4 ttl=128 time=0.446 ms
--- 192.168.1.165 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3058ms
rtt min/avg/max/mdev = 0.446/0.578/0.720/0.118 ms
(kali㉿kali)-[~]
$

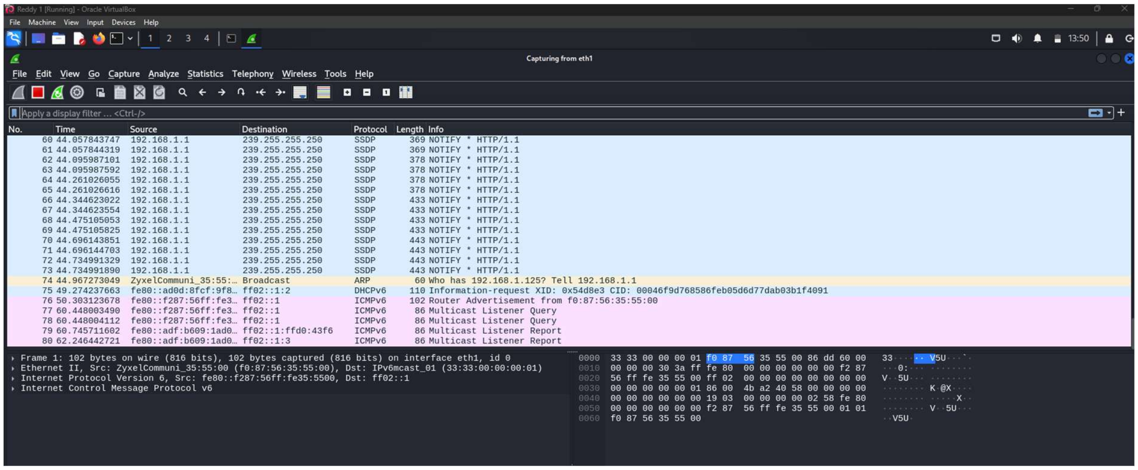


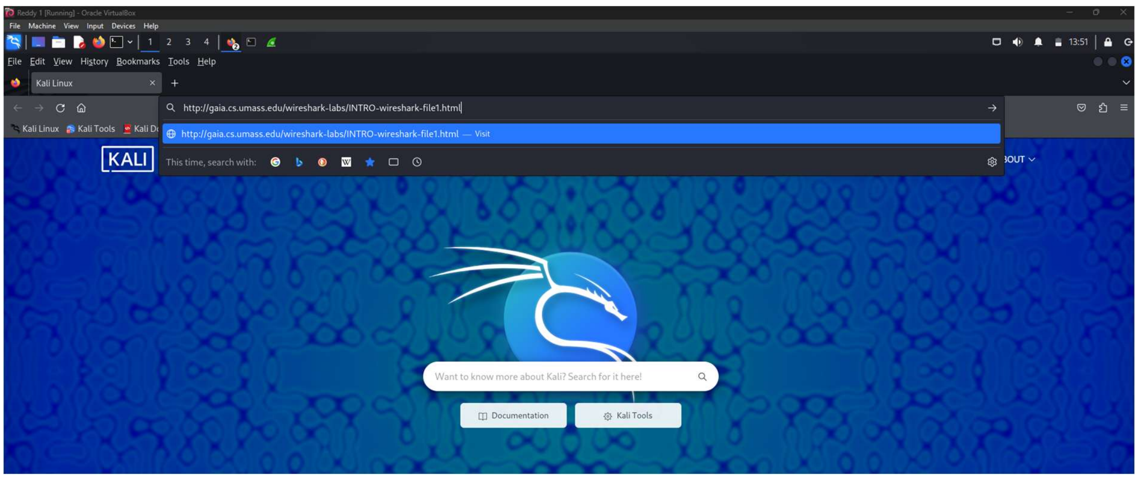
2. Check and prove from the Windows VM that Kali-Linux VM is up and running by using the appropriate command.

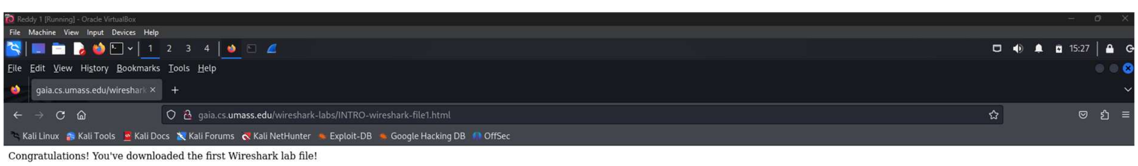


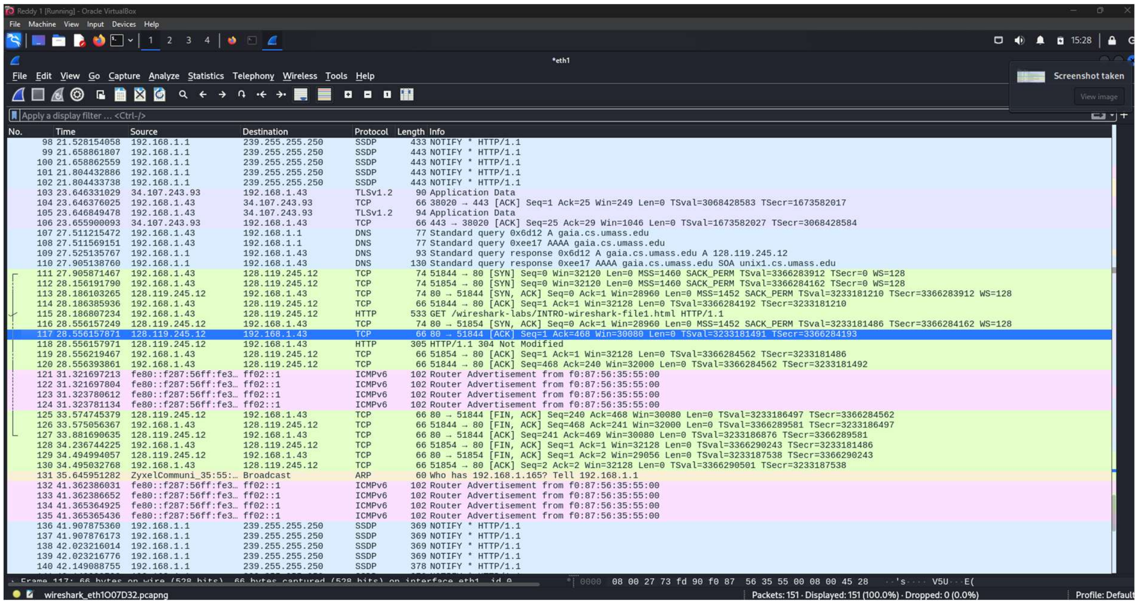

Reddy 2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Command Prompt
Microsoft Windows [Version 10.0.19045.2006]
(c) Microsoft Corporation. All rights reserved.
C:\Users\gregg>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=1ms TTL=64
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64
Reply from 192.168.1.3: bytes=32 time<1ms TTL=64
Reply from 192.168.1.3: bytes=32 time=1ms TTL=64
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\gregg>
Right Ctrl

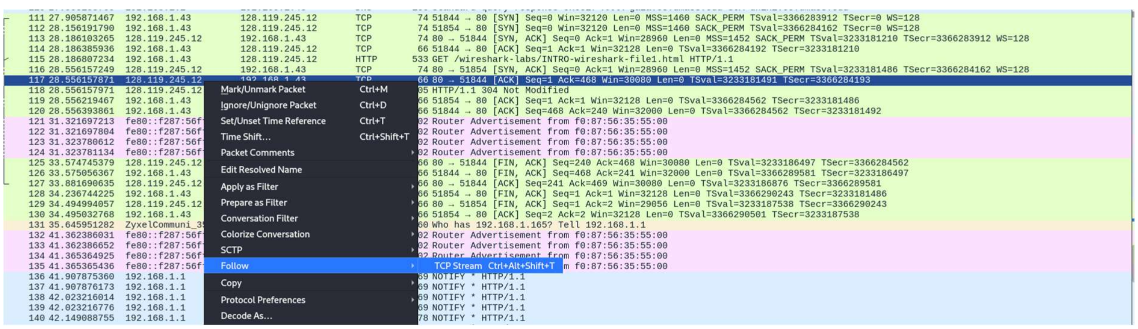

The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
Capture
...using this filter:
Enter a capture filter ...
All interfaces shown
eth0
eth1
any
Loopback: lo
bluetooth-monitor
nflog
nfqueue
dbus-system
dbus-session
Cisco remote capture: ciscodump
DisplayPort AUX channel monitor capture: dpauxmon
Random packet generator: randpkt
systemd Journal Export: sdjournal
SSH remote capture: sshdump
UDP Listener remote capture: udpdump
Wi-Fi remote capture: wifidump

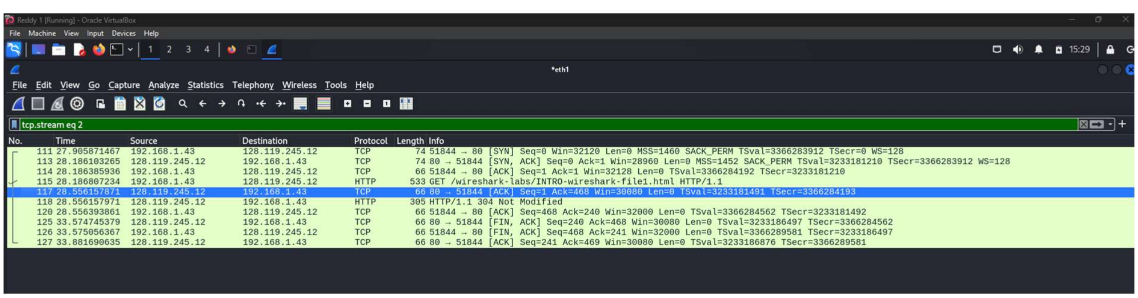

Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Capturing from eth1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter … <Ctrl-/>
No. Time Source Destination Protocol Length Info
60 44.057843747 192.168.1.1 239.255.255.250 SSDP 369 NOTIFY * HTTP/1.1
61 44.057844319 192.168.1.1 239.255.255.250 SSDP 369 NOTIFY * HTTP/1.1
62 44.095987101 192.168.1.1 239.255.255.250 SSDP 378 NOTIFY * HTTP/1.1
63 44.095987592 192.168.1.1 239.255.255.250 SSDP 378 NOTIFY * HTTP/1.1
64 44.261026055 192.168.1.1 239.255.255.250 SSDP 378 NOTIFY * HTTP/1.1
65 44.261026616 192.168.1.1 239.255.255.250 SSDP 378 NOTIFY * HTTP/1.1
66 44.344623022 192.168.1.1 239.255.255.250 SSDP 433 NOTIFY * HTTP/1.1
67 44.344623554 192.168.1.1 239.255.255.250 SSDP 433 NOTIFY * HTTP/1.1
68 44.475105053 192.168.1.1 239.255.255.250 SSDP 433 NOTIFY * HTTP/1.1
69 44.475105825 192.168.1.1 239.255.255.250 SSDP 433 NOTIFY * HTTP/1.1
70 44.696143851 192.168.1.1 239.255.255.250 SSDP 443 NOTIFY * HTTP/1.1
71 44.696144703 192.168.1.1 239.255.255.250 SSDP 443 NOTIFY * HTTP/1.1
72 44.734991329 192.168.1.1 239.255.255.250 SSDP 443 NOTIFY * HTTP/1.1
73 44.734991890 192.168.1.1 239.255.255.250 SSDP 443 NOTIFY * HTTP/1.1
74 44.967273049 ZyxelCommuni_35:55:… Broadcast ARP 60 Who has 192.168.1.125? Tell 192.168.1.1
75 49.274237663 fe80::ad0d:8fcf:9f8… ff02::1:2 DHCPv6 110 Information-request XID: 0x54d8e3 CID: 00046f9d768586feb05d6d77dab03b1f4091
76 50.303123678 fe80::f287:56ff:fe3… ff02::1 ICMPv6 102 Router Advertisement from f0:87:56:35:55:00
77 60.448003490 fe80::f287:56ff:fe3… ff02::1 ICMPv6 86 Multicast Listener Query
78 60.448004112 fe80::f287:56ff:fe3… ff02::1 ICMPv6 86 Multicast Listener Query
79 60.745711602 fe80::adf:b609:1ad0… ff02::1:ffd0:43f6 ICMPv6 86 Multicast Listener Report
80 62.246442721 fe80::adf:b609:1ad0… ff02::1:3 ICMPv6 86 Multicast Listener Report
Frame 1: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface eth1, id 0
Ethernet II, Src: ZyxelCommuni_35:55:00 (f0:87:56:35:55:00), Dst: IPv6mcast_01 (33:33:00:00:00:01)
Internet Protocol Version 6, Src: fe80::f287:56ff:fe35:5500, Dst: ff02::1
Internet Control Message Protocol v6
0000 33 33 00 00 00 01 f0 87 56 35 55 00 86 dd 60 00
0010 00 00 00 30 3a ff fe 80 00 00 00 00 00 00 f2 87
0020 56 ff fe 35 55 00 ff 02 00 00 00 00 00 00 00 00
0030 00 00 00 00 00 01 86 00 4b a2 40 58 00 00 00 00
0040 00 00 00 00 00 00 19 03 00 00 00 00 02 58 fe 80
0050 00 00 00 00 00 00 f2 87 56 ff fe 35 55 00 01 01
0060 f0 87 56 35 55 00


Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
File Edit View History Bookmarks Tools Help
Kali Linux
http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html
http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html — Visit
This time, search with:
KALI
Want to know more about Kali? Search for it here!
Documentation
Kali Tools


Reddy 1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
File Edit View History Bookmarks Tools Help
gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
Congratulations! You've downloaded the first Wireshark lab file!


Apply a display filter … <Ctrl-/>
Screenshot taken
wireshark_eth1OO7D32.pcapng
Packets: 151 · Displayed: 151 (100.0%) · Dropped: 0 (0.0%)
Profile: Default


Mark/Unmark Packet Ctrl+M
Ignore/Unignore Packet Ctrl+D
Set/Unset Time Reference Ctrl+T
Time Shift… Ctrl+Shift+T
Packet Comments
Edit Resolved Name
Apply as Filter
Prepare as Filter
Conversation Filter
Colorize Conversation
SCTP
Follow
TCP Stream Ctrl+Alt+Shift+T
Copy
Protocol Preferences
Decode As…


tcp.stream eq 2

- Packet 111-114 the TCP handshake takes place, packet 111 is the beginning of the connection and packet 114 is the completion of the handshake.
- Packet 115 is the HTTP request to the link inputted on mozila http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html
- Packet 117 was acknowledges the request
- Packet 118 is a response from the web server at this point it displays the web page

- Packet 125-127 TCP connection has ended in termination